Muhammad Asad

Principal Software Engineer | **Identity & Access Management** | **OAuth2 · OIDC · SSO** | **Workload & Machine-to-Machine Identity**

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PROFILE

Principal Software Engineer and Technical Lead with 7+ years building and scaling secure, high-availability systems, specialising in Identity & Access Management — SSO, OAuth2, OpenID Connect and fine-grained authorisation. Designed and shipped a production multi-tenant enterprise IAM platform with external identity-provider federation, serving multiple customer-facing products across three business units. Leads multi-team initiatives across 20+ engineers, works directly with security and compliance stakeholders, and mentors senior engineers on system design and risk.

CORE SKILLS

Identity & Access	IAM platform architecture · Single Sign-On (SSO) · OAuth2 · OpenID Connect (OIDC) · SAML · Role-Based Access Control (RBAC) · multi-tenancy & tenant isolation · identity-provider federation (BYO-IdP) · machine-to-machine (M2M) and workload identity · OAuth2 client credentials · service accounts · domain verification · approval workflows · Keycloak
Backend	Node.js (NestJS, Express.js) · TypeScript · GraphQL / Apollo · REST API design · TypeORM · PostgreSQL · Redis · Elasticsearch · microservices · event-driven and distributed systems · high-availability design
Platform	Docker · Kubernetes · Cloudflare Workers · CI/CD pipelines · observability · performance and scalability tuning · system design
AI / Gen AI	AI-assisted engineering workflows (GitHub Copilot, Claude / Claude Code) · AI-accelerated code review, refactoring and test generation · prompt design for developer tooling · automated signal-generation and analysis pipelines
Leadership	Technical leadership across teams · architecture governance and design review · mentoring and engineer development · stakeholder management with product, security and compliance · delivery standards
Frontend	Angular · single-page applications (SPA) · role-based admin interfaces
Current focus	Agentic identity — extending OAuth2 token exchange and scoped, short-lived credentials to AI agents · LLM API integration with structured output for news extraction and sentiment-based signal scoring in Cyrren

EXPERIENCE

Zones IT Solutions, LLC — Karachi, Pakistan — Aug 2022 – Present

Principal Software Engineer / Technical Lead · promoted from Senior Software Engineer, Nov 2023

- Technical lead for the enterprise Identity and Access Management platform underpinning multiple customer-facing products, and head of the System Integration team.
- Designed and owned a multi-tenant IAM platform providing tenant isolation across three business units, role-based access control, three-tier application visibility, self-service organisation onboarding, domain verification, bring-your-own identity-provider registration and multi-step approval workflows.
- Designed, delivered and ran in production external company federation, letting partner organisations authenticate through their own identity provider and share resources across tenant boundaries while preserving isolation and compliance.
- Provided machine-to-machine authentication for partner and internal systems — client-credentials access to platform APIs with no user interaction, governed by the same tenant isolation and role model.
- Made the long-term architectural decisions on tenant isolation, security boundaries, scalability and maintainability, delivered with no P1 identity incidents.
- Led and coordinated engineering across 20+ engineers in multiple teams, holding alignment on architecture and delivery standards.
- Translated regulatory, security and compliance requirements into practical technical design alongside product and security stakeholders.
- Introduced Claude Code and GitHub Copilot into the team's development workflow, and defined where AI-generated code is permitted and where it requires manual implementation, given the security sensitivity of authentication paths.
- Mentored senior and mid-level engineers through design reviews and architecture discussions, and contributed to a graduate rotational programme that trained roughly 50 engineers.

Airlift Technologies — Karachi, Pakistan — Jul 2019 – Jun 2022

Software Engineer (Backend / Full Stack)

- Backend and full-stack engineer through Airlift's growth from a mass-transit service into Airlift Express, a quick-commerce business delivering groceries in under 30 minutes — both running at high traffic with little tolerance for downtime.
- Designed and implemented microservices using Node.js, PostgreSQL and cloud-native patterns.
- Built internal shared libraries and npm packages, including authorisation guards reused across multiple services.
- Integrated AWS services — Lambda, EC2, S3, Aurora and CloudWatch — to support scalable, reliable deployments.
- Delivered features under time-critical, high-traffic conditions with cross-functional teams.

Experia — Karachi, Pakistan Dec 2018 – Jun 2019

Associate Software Engineer (Frontend)

- Developed and maintained single-page applications in Angular, ensuring performance on both mobile and desktop.
- Designed modular admin panels with role-based access control.
- Integrated real-time notifications and third-party APIs; coordinated delivery with design and backend.

SELECTED PROJECTS

Access Studio — Enterprise IAM Platform · Zones · Technical Lead

Multi-tenant authentication and authorisation platform delivering SSO and OAuth2/OIDC integration for partners and customers: tenant isolation across business units, RBAC, tiered application visibility, self-service onboarding, domain verification, BYO-IdP federation and approval workflows. Node.js / NestJS, PostgreSQL, Redis, Keycloak.

Cyrren · cyrren.io · Founder

Automated technical-intelligence platform for the Pakistan Stock Exchange — scheduled market-data pipelines, an automated technical-analysis engine and a subscriber-facing web product. Node.js / TypeScript, PostgreSQL.

EDUCATION & ADDITIONAL

BS, Computer Science Sep 2014 – Aug 2018
DHA Suffa University, Karachi, Pakistan

Languages	English (professional) · Urdu (native)
Citizenship	Pakistan
Availability	At short notice
References	Available on request